

# Daily Research Report: Dynamical Study of $s\bar{s} \rightarrow \Lambda\bar{\Lambda}$ Spin Correlations

Research synthesis – 23 August 2026

## Abstract

This report reconciles today's quantum-dynamical study of strange-quark pair spin correlations. The internally consistent picture is: the observable is a connected two-body spin tensor; relativistic axial tensors require momentum-dependent spin-tetrad projection; independent one-body spin relaxation damps the pair mode with the sum of leg rates; stochastic fields relax the isotropic scalar only through the relative field; finite-memory noise gives a non-Markovian kernel; and hadronization, feed-down and same-source dilution are indispensable.

## 1 Physical observable

Define  $C_{ab} = \langle \sigma_{1a} \sigma_{2b} \rangle - \langle \sigma_{1a} \rangle \langle \sigma_{2b} \rangle$  and  $P = \delta_{ab} C_{ab} / 3$ . For vanishing one-body polarization,  $P = \langle \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \rangle / 3$ . STAR reports  $P_{\Lambda\bar{\Lambda}} = 0.181 \pm 0.035_{\text{stat}} \pm 0.022_{\text{sys}}$  for short-range pairs and long-range consistency with zero; CMS at LHC energies finds no statistically significant  $\Lambda\bar{\Lambda}$  deviation from zero.

## 2 Density operator and BBGKY

Write  $\rho_{12} = \frac{1}{4}[I \otimes I + p_a \sigma_a \otimes I + \bar{p}_b I \otimes \sigma_b + C_{ab}^{\text{full}} \sigma_a \otimes \sigma_b]$ . Using  $\text{Tr}(\sigma_a \sigma_b) = 2\delta_{ab}$  recovers each coefficient and  $C_{ab} = C_{ab}^{\text{full}} - p_a \bar{p}_b$ . From  $i\partial_t \rho_N = [H_N, \rho_N]$ , tracing spectators gives  $i\partial_t \rho_{12} = [H_{12}, \rho_{12}] + \text{Tr}_3([V_{13} + V_{23}, \rho_{123}])$ , showing that closure requires a cluster/self-energy approximation.

## 3 Relativistic spin projection

For  $u\bar{u} = \frac{1}{2}(\not{p} + m)(1 + \gamma^5 \not{S})$ , gamma traces give  $\bar{u}\gamma^\mu\gamma^5 u = 2mS^\mu$ . Introduce  $e_a \cdot p = 0$ ,  $e_a \cdot e_b = -\delta_{ab}$  and  $\mathcal{A}^{\mu\nu} = e_{1a}^\mu e_{2b}^\nu C_{ab}$ . Hence  $C_{ab} = e_{1a\mu} e_{2b\nu} \mathcal{A}^{\mu\nu}$  and

$$P = \frac{1}{3} \sum_a e_{1a\mu} e_{2a\nu} \mathcal{A}^{\mu\nu}.$$

This corrects a naive lab-frame spatial trace.

## 4 Independent relaxation

For  $\mathcal{L}_m \rho = \frac{\gamma_m}{4} \sum_a (\sigma_{ma} \rho \sigma_{ma} - \rho)$  and  $\sum_a \sigma_a \sigma_i \sigma_a = -\sigma_i$ ,  $\mathcal{L}_m^\dagger \sigma_{mi} = -\gamma_m \sigma_{mi}$ . Therefore

$$\dot{P} = -(\gamma_s + \gamma_{\bar{s}})P, \quad P(t_h) = P(t_0)e^{-\int(\gamma_s + \gamma_{\bar{s}})dt}.$$

Perturbative strange-quark spin equilibration estimates are much slower than heavy-ion dynamical times.

## 5 Relative stochastic fields

Let  $H_I = \frac{1}{2}(\xi_1 \cdot \sigma_1 + \xi_2 \cdot \sigma_2)$ . With  $\bar{\xi} = (\xi_1 + \xi_2)/2$  and  $\xi_- = \xi_1 - \xi_2$ , the common part couples to  $\sigma_1 + \sigma_2$ . For  $Q = \sigma_1 \cdot \sigma_2$ ,  $[Q, \sigma_1 + \sigma_2] = 0$ , so perfectly common noise cannot change the scalar correlation. For  $\langle \xi_m^a(t) \xi_n^b(t') \rangle = 2D_{mn} \delta^{ab} \delta(t - t')$ ,

$$\dot{\rho} = -\frac{1}{4} \sum_{mn,a} D_{mn} [\sigma_m^a, [\sigma_n^a, \rho]],$$

and Pauli algebra yields

$$\dot{C}_{ij} = -2(D_{11} + D_{22})C_{ij} + 2D_{12}(\delta_{ij}C_{kk} - C_{ji}).$$

For  $C_{ij} = P\delta_{ij}$  and equivalent trajectories,

$$\boxed{\Gamma_{\text{rel}}(r) = 4[D(0) - D(r)], \quad \dot{P} = -\Gamma_{\text{rel}}P.}$$

## 6 Finite memory

With  $\langle \xi_m^a(t) \xi_n^b(t - \tau) \rangle = \delta^{ab} K_{mn}(\tau)$ , second-order TCL gives

$$\boxed{\dot{P}(t) = -2P(t) \int_0^t d\tau [K_{11} + K_{22} - K_{12} - K_{21}](\tau).}$$

For equivalent legs this is  $\dot{P} = -4P \int_0^t (K_0 - K_r) d\tau$ . If  $K_0 - K_r = K_\Delta e^{-\tau/\tau_c}$ ,

$$P(t) = P(0) e^{-4K_\Delta \tau_c [t - \tau_c (1 - e^{-t/\tau_c})]}.$$

Thus short-time decay is quadratic and long-time decay becomes exponential.

## 7 Gauge-covariant QCD interpretation

A candidate microscopic kernel is  $\mathcal{K}_{12}^{ij} \sim \kappa_s^2 \langle B^{ai}(x_1) \mathcal{U}_{C_{12}}^{ab} B^{bj}(x_2) \rangle$ . The Wilson line is required for gauge covariance. The coefficient  $\kappa_s$  must be matched from relativistic axial kinetic theory; importing a heavy-quark Pauli coefficient is not justified.

## 8 Kadanoff–Baym/Wigner route

Dyson equations imply  $(i \not{\partial} - m)G^< = \Sigma^R \circ G^< + \Sigma^< \circ G^A$  and its right-acting partner. With  $X = (x_1 + x_2)/2$ ,  $s = x_1 - x_2$ ,  $G^<(X, p) = \int d^4s e^{ip \cdot s} G^<(X + s/2, X - s/2)$ . Convolutions become Moyal products  $A \star B = A \exp[\frac{i}{2}(\overleftarrow{\partial}_X \cdot \overrightarrow{\partial}_p - \overleftarrow{\partial}_p \cdot \overrightarrow{\partial}_X)] B$ . Decompose  $W = \frac{1}{4}(\mathcal{F} + i\gamma^5 \mathcal{P} + \gamma^\mu \mathcal{V}_\mu + \gamma^\mu \gamma^5 \mathcal{A}_\mu + \frac{1}{2} \sigma^{\mu\nu} \mathcal{S}_{\mu\nu})$ . The pair problem requires the connected four-point function, double Wigner transform, axial–axial projection, and finally the tetrad contraction. The target structure is  $\mathcal{D}P = \mathcal{I}_{\text{rel}}[P] + \mathcal{S}_{\text{pair}} + \mathcal{R}_{\text{basis}}$ .

## 9 Hadronization

A minimal map is  $C_{ij}^{\Lambda\bar{\Lambda}} = \sum_{ab} T_{ia}^{(\Lambda)} \bar{T}_{jb}^{(\bar{\Lambda})} C_{ab}^{s\bar{s}} + C_{ij}^{\text{conn, had}}$ . With feed-down and source mixing,

$$\boxed{P_{\Lambda\bar{\Lambda}}^{\text{obs}} = D_{\text{had}} f_{\text{same}} P_{s\bar{s}}^{\text{src}} S_{\text{rel}} + P_{\text{conn}}^{\text{had/resc}}.}$$

This factorization is phenomenological and can fail if spin and kinematics are nonfactorizable.

## 10 Reconciled conclusions and next step

The surviving conclusions are: connected two-body tensors are essential; tetrad projection is mandatory; independent relaxation rates add; common-mode noise cancels exactly for the scalar; relative-field damping gives the white-noise and finite-memory limits; perturbative spin flips appear too slow to dominate; and hadronization/feed-down/source mixing are indispensable. The highest-priority next step is a gauge-covariant connected four-point Kadanoff–Baym derivation with explicit color/spinor indices and an axial–axial projection, followed by quantitative estimates of same-source fractions at RHIC and LHC energies.

## References

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